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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PROCESS DATA SHEET: VACUUM EJECTOR & LIQUID RING PACKAGE

Document Code: DS-X2301 | Revision: Z1 | Status: Issued for Construction

Equipment Tag: X-2301

Equipment Name: Preflash Column Vacuum Producing Package

Equipment Class: Two-Stage Steam Jet Ejector & Surface Condenser Package

Process Unit: Unit 2300 (Concentration & Decomposition Section)

Service: Maintains 35 mmHgA deep vacuum on Preflash Column V-2301 via Overhead Stream S220

1. PERFORMANCE & DESIGN DATA

- Design Suction Pressure: 35.0 mmHgA (0.046 kg/cm2a)
- Suction Temperature: 68.0 degC
- Motive Steam Supply: Medium Pressure (MP) Steam at 10.5 kg/cm2g and 195 degC (Consumption: 640 kg/hr)
- Inter-Condenser & After-Condenser Duty: 1.92 Gcal/hr using Cooling Water at 32 degC
- Wetted Material of Construction: SA-240 Type 316L Stainless Steel

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Equipment Tag: X-2301

- Vacuum Control Loop: Pressure Transmitter PT-0401 on V-2301 overhead modulates spillback control valve PV-0401
- Safety Criticality: Loss of vacuum in X-2301 causes V-2301 boiling temperature to rise above 105 degC, triggering

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Equipment Tag: X-2301

- 1st Stage Ejector EJ-2301A/B (100% redundant)
- Inter-Stage Surface Condenser EC-2301 (316L Tubes)
- 2nd Stage Liquid Ring Vacuum Pump VP-2301A/B (Cumene Seal Liquid)